

Lecture 9: HPC System Access, Job Submission, and SLURM

Objective:

To understand how to securely access High-Performance Computing (HPC) systems and manage job execution using SLURM workload manager.

Duration: 1 Hour

Learning Outcomes

By the end of this lecture, students will be able to:

- Access an HPC cluster using secure protocols
- Understand the difference between interactive and batch jobs
- Write and submit SLURM job scripts
- Use essential SLURM commands to monitor and manage jobs

1. HPC System Access

a. Accessing via SSH (Secure Shell)

SSH is the standard way to remotely access a server or HPC cluster.

Command:

```
ssh username@hpc.server.edu
```

Example:

```
ssh ramkumar@192.168.1.100
```

b. Environment Setup

Once connected:

- Load software using module load
- Set environment variables (export VAR=value)
- Work within user directories (/home/username/)

Example:

```
module load python/3.9
export OMP_NUM_THREADS=4
```

2. HPC Job Submission

a. Types of Jobs

Type	Description
Interactive Job	Immediate shell access to compute node
Batch Job	Scheduled to run using a job script

b. Submitting an Interactive Job

Use srun for direct interactive access:

```
srun --pty bash
```

c. Submitting a Batch Job using Job Script

Sample SLURM Job Script (job.sh):bash

```
#!/bin/bash
```

```
#SBATCH --job-name=myjob
```

```
#SBATCH --output=result.out
```

```
#SBATCH --ntasks=1
```

```
#SBATCH --time=00:10:00
```

```
#SBATCH --mem=1G
```

```
echo "Running my job on $(hostname)"
```

```
python3 myscript.py
```

Submit the job:

```
bash
```

```
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```

```
SBATCH job.sh
```

3. SLURM Workload Manager

SLURM (Simple Linux Utility for Resource Management) is used for:

- Job scheduling
- Resource allocation
- Monitoring and controlling tasks

a. Key SLURM Commands

Command	Description	Example
sbatch	Submit a batch job	sbatch job.sh
squeue	View job queue	squeue -u username
scancel	Cancel a job	scancel 12345
sinfo	View available partitions/nodes	sinfo

b. SLURM Directives Explained

Directive	Purpose
--job-name	Name of the job
--output	Output file for standard output
--ntasks	Number of parallel tasks

--time	Job run time limit (hh:mm:ss)
--mem	Memory requested

Classroom Activities

♦ Activity 1: SSH Practice (if lab available)

- Connect to a test server using SSH
- Navigate folders and create a directory

♦ Activity 2: Job Script Creation

- Write a job script that prints hostname and current date
- Submit using sbatch, and check with squeue

♦ Activity 3: Troubleshoot the Script

- Given a job script with a typo or missing directive, identify and fix the issue

Summary

Topic	Key Takeaways
SSH	Secure access to HPC systems
Interactive vs Batch Jobs	Immediate use vs. scheduled via scripts
SLURM	Most popular resource manager in HPC
sbatch, squeue, scancel	Used to submit, monitor, and cancel jobs

Homework / Assignment

Task:

Write and submit a SLURM job script that:

1. Loads a Python module



2. Runs a Python script that prints prime numbers from 1–100
3. Saves output in output.txt

Submission: Upload job.sh and output file before the next class